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Modeling Time Series with Changes in Regime

22.1. Introduction

Many variables undergo episodes in which the behavior of the series seems to change quite dramatically. A striking example is provided by Figure 22.1, which is taken from Rogers's (1992) study of the volume of dollar-denominated accounts held in Mexican banks. The Mexican government adopted various measures in 1982 to try to discourage the use of such accounts, and the effects are quite dramatic in a plot of the series.

Similar dramatic breaks will be seen if one follows almost any macroeconomic or financial time series for a sufficiently long period. Such apparent changes in the time series process can result from events such as wars, financial panics, or significant changes in government policies.

How should we model a change in the process followed by a particular time series? For the data plotted in Figure 22.1, one simple idea would be that the constant term for the autoregression changed in 1982. For data prior to 1982 we might use a model such as

$$y_t - \mu_1 = \phi(y_{t-1} - \mu_1) + \varepsilon_t, \quad [22.1.1]$$

while data after 1982 might be described by

$$y_t - \mu_2 = \phi(y_{t-1} - \mu_2) + \varepsilon_t, \quad [22.1.2]$$

where $\mu_2 < \mu_1$.

The specification in [22.1.1] and [22.1.2] seems a plausible description of the data in Figure 22.1, but it is not altogether satisfactory as a time series model. For example, how are we to forecast a series that is described by [22.1.1] and [22.1.2]? If the process has changed in the past, clearly it could also change again in the future, and this prospect should be taken into account in forming a forecast. Moreover, the change in regime surely should not be regarded as the outcome of a perfectly foreseeable, deterministic event. Rather, the change in regime is itself a random variable. A complete time series model would therefore include a description of the probability law governing the change from μ_1 to μ_2 .

These observations suggest that we might consider the process to be influenced by an unobserved random variable s_t^* , which will be called the *state* or *regime* that the process was in at date t . If $s_t^* = 1$, then the process is in regime 1, while $s_t^* = 2$ means that the process is in regime 2. Equations [22.1.1] and [22.1.2] can then equivalently be written as

$$y_t - \mu_{s_t^*} = \phi(y_{t-1} - \mu_{s_t^*}) + \varepsilon_t, \quad [22.1.3]$$

where $\mu_{s_t^*}$ indicates μ_1 when $s_t^* = 1$ and indicates μ_2 when $s_t^* = 2$.

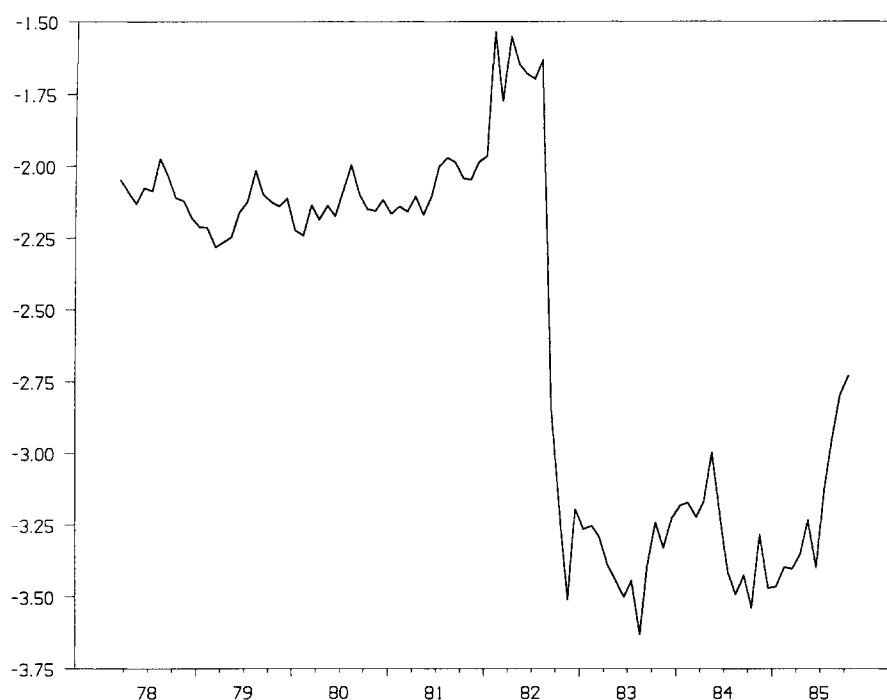


FIGURE 22.1 Log of the ratio of the peso value of dollar-denominated bank accounts in Mexico to the peso value of peso-denominated bank accounts in Mexico, monthly, 1978–85. (Rogers, 1992).

We then need a description of the time series process for the unobserved variable s_t^* . Since s_t^* takes on only discrete values (in this case, s_t^* is either 1 or 2), this will be a slightly different time series model from those for continuous-valued random variables considered elsewhere in this book.

The simplest time series model for a discrete-valued random variable is a *Markov chain*. The theory of Markov chains is reviewed in Section 22.2. In Section 22.4 this theory will be combined with a conventional time series model such as an autoregression that is assumed to characterize any given regime. Prior to doing so, however, it will be helpful to consider a special case of such processes, namely, that for which $\phi = 0$ in [22.1.3] and s_t^* is an i.i.d. discrete-valued random variable. Such a specification describes y_t as a simple mixture of different distributions, the statistical theory for which is reviewed in Section 22.3.

22.2. Markov Chains

Let s_t be a random variable that can assume only an integer value $\{1, 2, \dots, N\}$. Suppose that the probability that s_t equals some particular value j depends on the past only through the most recent value s_{t-1} :

$$P\{s_t = j | s_{t-1} = i, s_{t-2} = k, \dots\} = P\{s_t = j | s_{t-1} = i\} = p_{ij}. \quad [22.2.1]$$

Such a process is described as an N -state *Markov chain* with transition probabilities $\{p_{ij}\}_{i,j=1,2,\dots,N}$. The transition probability p_{ij} gives the probability that state i will be followed by state j . Note that

$$p_{i1} + p_{i2} + \dots + p_{iN} = 1. \quad [22.2.2]$$

It is often convenient to collect the transition probabilities in an $(N \times N)$ matrix $\mathbf{P}$ known as the *transition matrix*:

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{21} & \cdots & p_{N1} \\ p_{12} & p_{22} & \cdots & p_{N2} \\ \vdots & \vdots & \cdots & \vdots \\ p_{1N} & p_{2N} & \cdots & p_{NN} \end{bmatrix}. \quad [22.2.3]$$

The row j , column i element of $\mathbf{P}$ is the transition probability p_{ij} ; for example, the row 2, column 1 element gives the probability that state 1 will be followed by state 2.

Representing a Markov Chain with a Vector Autoregression

A useful representation for a Markov chain is obtained by letting ξ_t denote a random $(N \times 1)$ vector whose j th element is equal to unity if $s_t = j$ and whose j th element equals zero otherwise. Thus, when $s_t = 1$, the vector ξ_t is equal to the first column of $\mathbf{I}_N$ (the $N \times N$ identity matrix); when $s_t = 2$, the vector ξ_t is the second column of $\mathbf{I}_N$; and so on:

$$\xi_t = \begin{cases} (1, 0, 0, \dots, 0)' & \text{when } s_t = 1 \\ (0, 1, 0, \dots, 0)' & \text{when } s_t = 2 \\ \vdots & \vdots \\ (0, 0, 0, \dots, 1)' & \text{when } s_t = N. \end{cases}$$

If $s_t = i$, then the j th element of ξ_{t+1} is a random variable that takes on the value unity with probability p_{ij} and takes on the value zero otherwise. Such a random variable has expectation p_{ij} . Thus, the conditional expectation of ξ_{t+1} given $s_t = i$ is given by

$$E(\xi_{t+1} | s_t = i) = \begin{bmatrix} p_{i1} \\ p_{i2} \\ \vdots \\ p_{iN} \end{bmatrix}. \quad [22.2.4]$$

This vector is simply the i th column of the matrix $\mathbf{P}$ in [22.2.3]. Moreover, when $s_t = i$, the vector ξ_t corresponds to the i th column of $\mathbf{I}_N$, in which case the vector in [22.2.4] could be described as $\mathbf{P}\xi_t$. Hence, expression [22.2.4] implies that

$$E(\xi_{t+1} | \xi_t) = \mathbf{P}\xi_t,$$

and indeed, from the Markov property [22.2.1], it follows further that

$$E(\xi_{t+1} | \xi_t, \xi_{t-1}, \dots) = \mathbf{P}\xi_t. \quad [22.2.5]$$

Result [22.2.5] implies that it is possible to express a Markov chain in the form

$$\xi_{t+1} = \mathbf{P}\xi_t + \mathbf{v}_{t+1}, \quad [22.2.6]$$

where

$$\mathbf{v}_{t+1} \equiv \xi_{t+1} - E(\xi_{t+1} | \xi_t, \xi_{t-1}, \dots). \quad [22.2.7]$$

Expression [22.2.6] has the form of a first-order vector autoregression for ξ_t ; note that [22.2.7] implies that the innovation $\mathbf{v}_t$ is a martingale difference sequence. Although the vector $\mathbf{v}_t$ can take on only a finite set of values, on average $\mathbf{v}_t$ is zero. Moreover, the value of $\mathbf{v}_t$ is impossible to forecast on the basis of previous states of the process.

Forecasts for a Markov Chain

Expression [22.2.6] implies that

$$\xi_{t+m} = \mathbf{v}_{t+m} + \mathbf{P}\mathbf{v}_{t+m-1} + \mathbf{P}^2\mathbf{v}_{t+m-2} + \cdots + \mathbf{P}^{m-1}\mathbf{v}_{t+1} + \mathbf{P}^m\xi_t, \quad [22.2.8]$$

where $\mathbf{P}^m$ indicates the transition matrix multiplied by itself m times. It follows from [22.2.8] that m -period-ahead forecasts for a Markov chain can be calculated from

$$E(\xi_{t+m} | \xi_t, \xi_{t-1}, \dots) = \mathbf{P}^m \xi_t. \quad [22.2.9]$$

Again, since the j th element of ξ_{t+m} will be unity if $s_{t+m} = j$ and zero otherwise, the j th element of the $(N \times 1)$ vector $E(\xi_{t+m} | \xi_t, \xi_{t-1}, \dots)$ indicates the probability that s_{t+m} takes on the value j , conditional on the state of the system at date t . For example, if the process is in state i at date t , then [22.2.9] asserts that

$$\begin{bmatrix} P\{s_{t+m} = 1 | s_t = i\} \\ P\{s_{t+m} = 2 | s_t = i\} \\ \vdots \\ P\{s_{t+m} = N | s_t = i\} \end{bmatrix} = \mathbf{P}^m \cdot \mathbf{e}_i, \quad [22.2.10]$$

where $\mathbf{e}_i$ denotes the i th column of $\mathbf{I}_N$. Expression [22.2.10] indicates that the m -period-ahead transition probabilities for a Markov chain can be calculated by multiplying the matrix $\mathbf{P}$ by itself m times. Specifically, the probability that an observation from regime i will be followed m periods later by an observation from regime j , $P\{s_{t+m} = j | s_t = i\}$, is given by the row j , column i element of the matrix $\mathbf{P}^m$.

Reducible Markov Chains

For a two-state Markov chain, the transition matrix is

$$\mathbf{P} = \begin{bmatrix} p_{11} & 1 - p_{22} \\ 1 - p_{11} & p_{22} \end{bmatrix}. \quad [22.2.11]$$

Suppose that $p_{11} = 1$, so that the matrix $\mathbf{P}$ is upper triangular. Then, once the process enters state 1, there is no possibility of ever returning to state 2. In such a case we would say that state 1 is an *absorbing state* and that the Markov chain is *reducible*.

More generally, an N -state Markov chain is said to be *reducible* if there exists a way to label the states (that is, a way to choose which state to call state 1, which to call state 2, and so on) such that the transition matrix can be written in the form

$$\mathbf{P} = \begin{bmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{0} & \mathbf{D} \end{bmatrix},$$

where $\mathbf{B}$ denotes a $(K \times K)$ matrix for some $1 \leq K < N$. If $\mathbf{P}$ is upper block-triangular, then so is $\mathbf{P}^m$ for any m . Hence, once such a process enters a state j such that $j \leq K$, there is no possibility of ever returning to one of the states $K + 1, K + 2, \dots, N$.

A Markov chain that is not reducible is said to be *irreducible*. For example, a two-state chain is irreducible if $p_{11} < 1$ and $p_{22} < 1$.

Ergodic Markov Chains

Equation [22.2.2] requires that every column of $\mathbf{P}$ sum to unity, or

$$\mathbf{P}'\mathbf{1} = \mathbf{1}, \quad [22.2.12]$$

where $\mathbf{1}$ denotes an $(N \times 1)$ vector of 1s. Expression [22.2.12] implies that unity is an eigenvalue of the matrix $\mathbf{P}'$ and that $\mathbf{1}$ is the associated eigenvector. Since a matrix and its transpose share the same eigenvalues, it follows that unity is an eigenvalue of the transition matrix $\mathbf{P}$ for any Markov chain.

Consider an N -state irreducible Markov chain with transition matrix $\mathbf{P}$. Suppose that one of the eigenvalues of $\mathbf{P}$ is unity and that all other eigenvalues of $\mathbf{P}$ are inside the unit circle. Then the Markov chain is said to be *ergodic*. The $(N \times 1)$ vector of *ergodic probabilities* for an ergodic chain is denoted $\boldsymbol{\pi}$. This vector $\boldsymbol{\pi}$ is defined as the eigenvector of $\mathbf{P}$ associated with the unit eigenvalue; that is, the vector of ergodic probabilities $\boldsymbol{\pi}$ satisfies

$$\mathbf{P}\boldsymbol{\pi} = \boldsymbol{\pi}. \quad [22.2.13]$$

The eigenvector $\boldsymbol{\pi}$ is normalized so that its elements sum to unity ($\mathbf{1}'\boldsymbol{\pi} = 1$). It can be shown that if $\mathbf{P}$ is the transition matrix for an ergodic Markov chain, then

$$\lim_{m \rightarrow \infty} \mathbf{P}^m = \boldsymbol{\pi} \cdot \mathbf{1}'. \quad [22.2.14]$$

We establish [22.2.14] here for the case when all the eigenvalues of $\mathbf{P}$ are distinct; a related argument based on the Jordan decomposition that is valid for ergodic chains with repeated eigenvalues is developed in Cox and Miller (1965, pp. 120–23). For the case of distinct eigenvalues, we know from [A.4.24] that $\mathbf{P}$ can always be written in the form

$$\mathbf{P} = \mathbf{T}\boldsymbol{\Lambda}\mathbf{T}^{-1}, \quad [22.2.15]$$

where $\mathbf{T}$ is an $(N \times N)$ matrix whose columns are the eigenvectors of $\mathbf{P}$ while $\boldsymbol{\Lambda}$ is a diagonal matrix whose diagonal contains the corresponding eigenvalues of $\mathbf{P}$. It follows as in [1.2.19] that

$$\mathbf{P}^m = \mathbf{T}\boldsymbol{\Lambda}^m\mathbf{T}^{-1}. \quad [22.2.16]$$

Since the $(1, 1)$ element of $\boldsymbol{\Lambda}$ is unity and all other elements of $\boldsymbol{\Lambda}$ are inside the unit circle, $\boldsymbol{\Lambda}^m$ converges to a matrix with unity in the $(1, 1)$ position and zeros elsewhere. Hence,

$$\lim_{m \rightarrow \infty} \mathbf{P}^m = \mathbf{x} \cdot \mathbf{y}', \quad [22.2.17]$$

where $\mathbf{x}$ is the first column of $\mathbf{T}$ and $\mathbf{y}'$ is the first row of $\mathbf{T}^{-1}$.

The first column of $\mathbf{T}$ is the eigenvector of $\mathbf{P}$ corresponding to the unit eigenvalue, which eigenvector was denoted $\boldsymbol{\pi}$ in [22.2.13]:

$$\mathbf{x} = \boldsymbol{\pi}. \quad [22.2.18]$$

Moreover, the first row of $\mathbf{T}^{-1}$, when expressed as a column vector, corresponds to the eigenvector of $\mathbf{P}'$ associated with the unit eigenvalue, which eigenvector was seen to be proportional to the vector $\mathbf{1}$ in [22.2.12]:

$$\mathbf{y} = \alpha \cdot \mathbf{1}. \quad [22.2.19]$$

To verify [22.2.19], note from [22.2.15] that the matrix of eigenvectors $\mathbf{T}$ of the matrix $\mathbf{P}$ is characterized by

$$\mathbf{P}\mathbf{T} = \mathbf{T}\boldsymbol{\Lambda}. \quad [22.2.20]$$

Transposing [22.2.15] results in

$$\mathbf{P}' = (\mathbf{T}^{-1})' \mathbf{A} \mathbf{T}',$$

and postmultiplying by $(\mathbf{T}^{-1})'$ yields

$$\mathbf{P}'(\mathbf{T}^{-1})' = (\mathbf{T}^{-1})' \mathbf{A}. \quad [22.2.21]$$

Comparing [22.2.21] with [22.2.20] confirms that the columns of $(\mathbf{T}^{-1})'$ correspond to eigenvectors of $\mathbf{P}'$. In particular, then, the first column of $(\mathbf{T}^{-1})'$ is proportional to the eigenvector of $\mathbf{P}'$ associated with the unit eigenvalue, which eigenvector was seen to be given by $\mathbf{1}$ in equation [22.2.12]. Since $\mathbf{y}$ was defined as the first column of $(\mathbf{T}^{-1})'$, this establishes the claim made in equation [22.2.19].

Substituting [22.2.18] and [22.2.19] into [22.2.17], it follows that

$$\lim_{m \rightarrow \infty} \mathbf{P}^m = \boldsymbol{\pi} \cdot \alpha \mathbf{1}'.$$

Since $\mathbf{P}^m$ can be interpreted as a matrix of transition probabilities, each column must sum to unity. Thus, since the vector of ergodic probabilities $\boldsymbol{\pi}$ was normalized by the condition that $\mathbf{1}'\boldsymbol{\pi} = 1$, it follows that the normalizing constant α must be unity, establishing the claim made in [22.2.14].

Result [22.2.14] implies that the long-run forecast for an ergodic Markov chain is independent of the current state, since, from [22.2.9],

$$E(\boldsymbol{\xi}_{t+m} | \boldsymbol{\xi}_t, \boldsymbol{\xi}_{t-1}, \dots) = \mathbf{P}^m \boldsymbol{\xi}_t \xrightarrow{P} \boldsymbol{\pi} \cdot \mathbf{1}' \boldsymbol{\xi}_t = \boldsymbol{\pi},$$

where the final equality follows from the observation that $\mathbf{1}'\boldsymbol{\xi}_t = 1$ regardless of the value of $\boldsymbol{\xi}_t$. The long-run forecast of $\boldsymbol{\xi}_{t+m}$ is given by the vector of ergodic probabilities $\boldsymbol{\pi}$ regardless of the current value of $\boldsymbol{\xi}_t$.

The vector of ergodic probabilities can also be viewed as indicating the unconditional probability of each of the N different states. To see this, suppose that we had used the symbol π_j to indicate the unconditional probability $P\{s_t = j\}$. Then the vector $\boldsymbol{\pi} \equiv (\pi_1, \pi_2, \dots, \pi_N)'$ could be described as the unconditional expectation of $\boldsymbol{\xi}_t$:

$$\boldsymbol{\pi} = E(\boldsymbol{\xi}_t). \quad [22.2.22]$$

If one takes unconditional expectations of [22.2.6], the result is

$$E(\boldsymbol{\xi}_{t+1}) = \mathbf{P} \cdot E(\boldsymbol{\xi}_t).$$

Assuming stationarity and using the definition [22.2.22], this becomes

$$\boldsymbol{\pi} = \mathbf{P} \cdot \boldsymbol{\pi},$$

which is identical to equation [22.2.13] characterizing $\boldsymbol{\pi}$ as the eigenvector of $\mathbf{P}$ associated with the unit eigenvalue. For an ergodic Markov chain, this eigenvector is unique, and so the vector of ergodic probabilities $\boldsymbol{\pi}$ can be interpreted as the vector of unconditional probabilities.

An ergodic Markov chain is a covariance-stationary process. Yet [22.2.6] takes the form of a VAR with a unit root, since one of the eigenvalues of $\mathbf{P}$ is unity. This VAR is stationary despite the unit root because the variance-covariance matrix of $\mathbf{v}_t$ is singular. In particular, since $\mathbf{1}'\boldsymbol{\xi}_t = 1$ for all t and since $\mathbf{1}'\mathbf{P} = \mathbf{1}'$, equation [22.2.6] implies that $\mathbf{1}'\mathbf{v}_t = 0$ for all t . Thus, from [22.2.19], the first element of the $(N \times 1)$ vector $\mathbf{T}^{-1}\mathbf{v}_t$ is always zero, meaning that from [22.2.16] the unit eigenvalue in $\mathbf{P}^m \mathbf{v}_t$ always has a coefficient of zero.

Further Discussion of Two-State Markov Chains

The eigenvalues of the transition matrix $\mathbf{P}$ for any N -state Markov chain are found from the solutions to $|\mathbf{P} - \lambda \mathbf{I}_N| = 0$. For the two-state Markov chain, the eigenvalues satisfy

$$\begin{aligned} 0 &= \begin{vmatrix} p_{11} - \lambda & 1 - p_{22} \\ 1 - p_{11} & p_{22} - \lambda \end{vmatrix} \\ &= (p_{11} - \lambda)(p_{22} - \lambda) - (1 - p_{11})(1 - p_{22}) \\ &= p_{11}p_{22} - (p_{11} + p_{22})\lambda + \lambda^2 - 1 + p_{11} + p_{22} - p_{11}p_{22} \\ &= \lambda^2 - (p_{11} + p_{22})\lambda - 1 + p_{11} + p_{22} \\ &= (\lambda - 1)(\lambda + 1 - p_{11} - p_{22}). \end{aligned}$$

Thus, the eigenvalues for a two-state chain are given by $\lambda_1 = 1$ and $\lambda_2 = -1 + p_{11} + p_{22}$. The second eigenvalue, λ_2 , will be inside the unit circle as long as $0 < p_{11} + p_{22} < 2$. We saw earlier that this chain is irreducible as long as $p_{11} < 1$ and $p_{22} < 1$. Thus, a two-state Markov chain is ergodic provided that $p_{11} < 1$, $p_{22} < 1$, and $p_{11} + p_{22} > 0$.

The eigenvector associated with λ_1 for the two-state chain turns out to be

$$\boldsymbol{\pi} = \begin{bmatrix} (1 - p_{22})/(2 - p_{11} - p_{22}) \\ (1 - p_{11})/(2 - p_{11} - p_{22}) \end{bmatrix}$$

(the reader is invited to confirm this and the claims that follow in Exercise 22.1). Thus, the unconditional probability that the process will be in regime 1 at any given date is given by

$$P\{s_t = 1\} = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}.$$

The unconditional probability that the process will be in regime 2, the second element of $\boldsymbol{\pi}$, is readily seen to be 1 minus this magnitude. The eigenvector associated with λ_2 is

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Thus, from [22.2.16], the matrix of m -period-ahead transition probabilities for an ergodic two-state Markov chain is given by

$$\begin{aligned} \mathbf{P}^m &= \begin{bmatrix} \frac{1 - p_{22}}{2 - p_{11} - p_{22}} & -1 \\ \frac{1 - p_{11}}{2 - p_{11} - p_{22}} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \lambda_2^m \end{bmatrix} \begin{bmatrix} 1 & 1 \\ \frac{-(1 - p_{11})}{2 - p_{11} - p_{22}} & \frac{1 - p_{22}}{2 - p_{11} - p_{22}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{(1 - p_{22}) + \lambda_2^m(1 - p_{11})}{2 - p_{11} - p_{22}} & \frac{(1 - p_{22}) - \lambda_2^m(1 - p_{22})}{(2 - p_{11} - p_{22})} \\ \frac{(1 - p_{11}) - \lambda_2^m(1 - p_{11})}{2 - p_{11} - p_{22}} & \frac{(1 - p_{11}) + \lambda_2^m(1 - p_{22})}{2 - p_{11} - p_{22}} \end{bmatrix}. \end{aligned}$$

Thus, for example if the process is currently in state 1, the probability that m periods later it will be in state 2 is given by

$$P\{s_{t+m} = 2 | s_t = 1\} = \frac{(1 - p_{11}) - \lambda_2^m(1 - p_{11})}{2 - p_{11} - p_{22}},$$

where $\lambda_2 = -1 + p_{11} + p_{22}$.

A two-state Markov chain can also be represented by a simple scalar $AR(1)$ process, as follows. Let ξ_{1t} denote the first element of the vector ξ_t ; that is, ξ_{1t} is a random variable that is equal to unity when $s_t = 1$ and equal to zero otherwise. For the two-state chain, the second element of ξ_t is then $1 - \xi_{1t}$. Hence, [22.2.6] can be written

$$\begin{bmatrix} \xi_{1,t+1} \\ 1 - \xi_{1,t+1} \end{bmatrix} = \begin{bmatrix} p_{11} & 1 - p_{22} \\ 1 - p_{11} & p_{22} \end{bmatrix} \begin{bmatrix} \xi_{1t} \\ 1 - \xi_{1t} \end{bmatrix} + \begin{bmatrix} v_{1,t+1} \\ v_{2,t+1} \end{bmatrix}. \quad [22.2.23]$$

The first row of [22.2.23] states that

$$\xi_{1,t+1} = (1 - p_{22}) + (-1 + p_{11} + p_{22})\xi_{1t} + v_{1,t+1}. \quad [22.2.24]$$

Expression [22.2.24] will be recognized as an $AR(1)$ process with constant term $(1 - p_{22})$ and autoregressive coefficient equal to $(-1 + p_{11} + p_{22})$. Note that this autoregressive coefficient turns out to be the second eigenvalue λ_2 of $\mathbf{P}$ calculated previously. When $p_{11} + p_{22} > 1$, the process is likely to persist in its current state and the variable ξ_{1t} would be positively serially correlated, whereas when $p_{11} + p_{22} < 1$, the process is more likely to switch out of a state than stay in it, producing negative serial correlation. Recall further from equation [3.4.3] that the mean of a first-order autoregression is given by $c/(1 - \phi)$. Hence, the representation [22.2.24] implies that

$$E(\xi_{1t}) = \frac{1 - p_{22}}{2 - p_{11} - p_{22}},$$

which reproduces the earlier calculation of the value for the ergodic probability π_1 .

Calculating Ergodic Probabilities for an N -state Markov Chain

For a general ergodic N -state process, the vector of unconditional probabilities represents a vector $\boldsymbol{\pi}$ with the properties that $\mathbf{P}\boldsymbol{\pi} = \boldsymbol{\pi}$ and $\mathbf{1}'\boldsymbol{\pi} = 1$, where $\mathbf{1}$ denotes an $(N \times 1)$ vector of 1s. We thus seek a vector $\boldsymbol{\pi}$ satisfying

$$\mathbf{A}\boldsymbol{\pi} = \mathbf{e}_{N+1}. \quad [22.2.25]$$

where $\mathbf{e}_{N+1}$ denotes the $(N + 1)$ th column of $\mathbf{I}_{N+1}$ and where

$$\mathbf{A}_{(N+1) \times N} = \begin{bmatrix} \mathbf{I}_N - \mathbf{P} \\ \mathbf{1}' \end{bmatrix}.$$

Such a solution can be found by premultiplying [22.2.25] by $(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'$:

$$\boldsymbol{\pi} = (\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{e}_{N+1}. \quad [22.2.26]$$

In other words, $\boldsymbol{\pi}$ is the $(N + 1)$ th column of the matrix $(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'$.

Periodic Markov Chains

If a Markov chain is irreducible, then there is one and only one eigenvalue equal to unity. However, there may be more than one eigenvalue on the unit circle, meaning that not all irreducible Markov chains are ergodic. For example, consider a two-state Markov chain in which $p_{11} = p_{22} = 0$:

$$\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The eigenvalues of this transition matrix are $\lambda_1 = 1$ and $\lambda_2 = -1$, both of which are on the unit circle. Thus, the matrix $\mathbf{P}^m$ does not converge to any fixed limit of the form $\boldsymbol{\pi} \cdot \mathbf{1}'$ for this case. Instead, if the process is in state 1 at date t , then it is certain to be there again for dates $t + 2, t + 4, t + 6, \dots$, with no tendency to converge as $m \rightarrow \infty$. Such a Markov chain is said to be *periodic* with period 2.

In general, it is possible to show that for any irreducible N -state Markov chain, all the eigenvalues of the transition matrix will be on or inside the unit circle. If there are K eigenvalues strictly on the unit circle with $K > 1$, then the chain is said to be periodic with period K . Such chains have the property that the states can be classified into K distinct classes, such that if the state at date t is from class α , then the state at date $t + 1$ is certain to be from class $\alpha + 1$ (where class $\alpha + 1$ for $\alpha = K$ is interpreted to be class 1). Thus, there is a zero probability of returning to the original state s_t , and indeed zero probability of returning to any member of the original class α , except at horizons that are integer multiples of the period (such as dates $t + K, t + 2K, t + 3K$, and so on). For further discussion of periodic Markov chains, see Cox and Miller (1965).

22.3. Statistical Analysis of i.i.d. Mixture Distributions

In Section 22.4, we will consider autoregressive processes in which the parameters of the autoregression can change as the result of a regime-shift variable. The regime itself will be described as the outcome of an unobserved Markov chain. Before analyzing such processes, it is instructive first to consider a special case of these processes known as *i.i.d. mixture distributions*.

Let the regime that a given process is in at date t be indexed by an unobserved random variable s_t , where there are N possible regimes ($s_t = 1, 2, \dots$, or N). When the process is in regime 1, the observed variable y_t is presumed to have been drawn from a $N(\mu_1, \sigma_1^2)$ distribution. If the process is in regime 2, then y_t is drawn from a $N(\mu_2, \sigma_2^2)$ distribution, and so on. Hence, the density of y_t conditional on the random variable s_t taking on the value j is

$$f(y_t | s_t = j; \boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left\{-\frac{(y_t - \mu_j)^2}{2\sigma_j^2}\right\} \quad [22.3.1]$$

for $j = 1, 2, \dots, N$. Here $\boldsymbol{\theta}$ is a vector of population parameters that includes $\mu_1, \dots, \mu_N$ and $\sigma_1^2, \dots, \sigma_N^2$.

The unobserved regime $\{s_t\}$ is presumed to have been generated by some probability distribution, for which the unconditional probability that s_t takes on the value j is denoted π_j :

$$P\{s_t = j; \boldsymbol{\theta}\} = \pi_j \quad \text{for } j = 1, 2, \dots, N. \quad [22.3.2]$$

The probabilities $\pi_1, \dots, \pi_N$ are also included in θ ; that is, θ is given by

$$\theta \equiv (\mu_1, \dots, \mu_N, \sigma_1^2, \dots, \sigma_N^2, \pi_1, \dots, \pi_N)'.$$

Recall that for any events A and B , the conditional probability of A given B is defined as

$$P\{A|B\} = \frac{P\{A \text{ and } B\}}{P\{B\}},$$

assuming that the probability that event B occurs is not zero. This expression implies that the joint probability of A and B occurring together can be calculated as

$$P\{A \text{ and } B\} = P\{A|B\} \cdot P\{B\}.$$

For example, if we were interested in the probability of the joint event that $s_t = j$ and that y_t falls within some interval $[c, d]$, this could be found by integrating

$$p(y_t, s_t = j; \theta) = f(y_t | s_t = j; \theta) \cdot P\{s_t = j; \theta\} \quad [22.3.3]$$

over all values of y_t between c and d . Expression [22.3.3] will be called the *joint density-distribution function* of y_t and s_t . From [22.3.1] and [22.3.2], this function is given by

$$p(y_t, s_t = j; \theta) = \frac{\pi_j}{\sqrt{2\pi}\sigma_j} \exp\left\{-\frac{(y_t - \mu_j)^2}{2\sigma_j^2}\right\}. \quad [22.3.4]$$

The unconditional density of y_t can be found by summing [22.3.4] over all possible values for j :

$$\begin{aligned} f(y_t; \theta) &= \sum_{j=1}^N p(y_t, s_t = j; \theta) \\ &= \frac{\pi_1}{\sqrt{2\pi}\sigma_1} \exp\left\{-\frac{(y_t - \mu_1)^2}{2\sigma_1^2}\right\} \\ &\quad + \frac{\pi_2}{\sqrt{2\pi}\sigma_2} \exp\left\{-\frac{(y_t - \mu_2)^2}{2\sigma_2^2}\right\} + \dots \\ &\quad + \frac{\pi_N}{\sqrt{2\pi}\sigma_N} \exp\left\{-\frac{(y_t - \mu_N)^2}{2\sigma_N^2}\right\}. \end{aligned} \quad [22.3.5]$$

Since the regime s_t is unobserved, expression [22.3.5] is the relevant density describing the actually observed data y_t . If the regime variable s_t is distributed i.i.d. across different dates t , then the log likelihood for the observed data can be calculated from [22.3.5] as

$$\mathcal{L}(\theta) = \sum_{t=1}^T \log f(y_t; \theta). \quad [22.3.6]$$

The maximum likelihood estimate of θ is obtained by maximizing [22.3.6] subject to the constraints that $\pi_1 + \pi_2 + \dots + \pi_N = 1$ and $\pi_j \geq 0$ for $j = 1, 2, \dots, N$. This can be achieved using the numerical methods described in Section 5.7, or using the *EM* algorithm developed later in this section.

Functions of the form of [22.3.5] can be used to represent a broad class of different densities. Figure 22.2 gives an example for $N = 2$. The joint density-distribution $p(y_t, s_t = 1; \theta)$ is π_1 times a $N(\mu_1, \sigma_1^2)$ density, while $p(y_t, s_t = 2; \theta)$ is π_2 times a $N(\mu_2, \sigma_2^2)$ density. The unconditional density for the observed variable $f(y_t; \theta)$ is the sum of these two magnitudes.

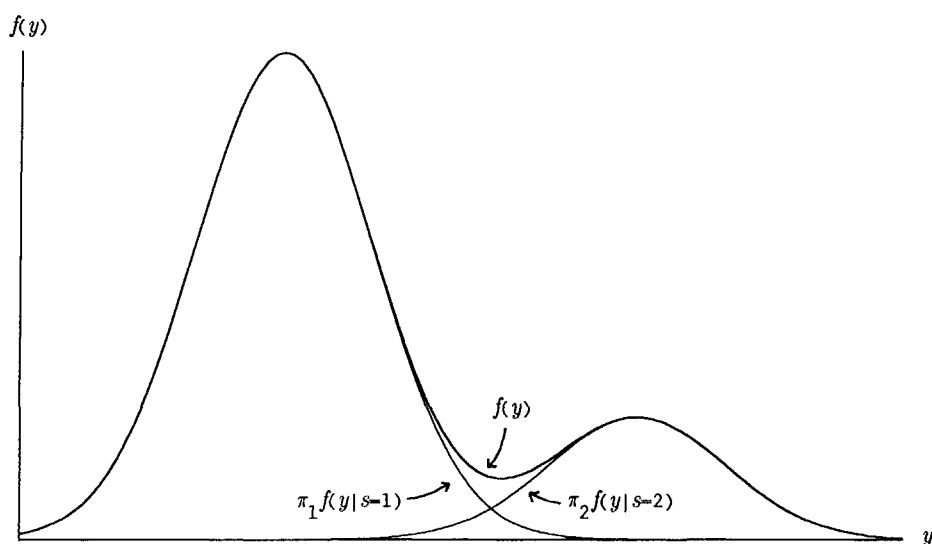


FIGURE 22.2 Density of mixture of two Gaussian distributions with $y_t|s_t = 1 \sim N(0, 1)$, $y_t|s_t = 2 \sim N(4, 1)$, and $P\{s_t = 1\} = 0.8$.

A mixture of two Gaussian variables need not have the bimodal appearance of Figure 22.2. Gaussian mixtures can also produce a unimodal density, allowing skew or kurtosis different from that of a single Gaussian variable, as in Figure 22.3.

Inference About the Unobserved Regime

Once one has obtained estimates of θ , it is possible to make an inference about which regime was more likely to have been responsible for producing the

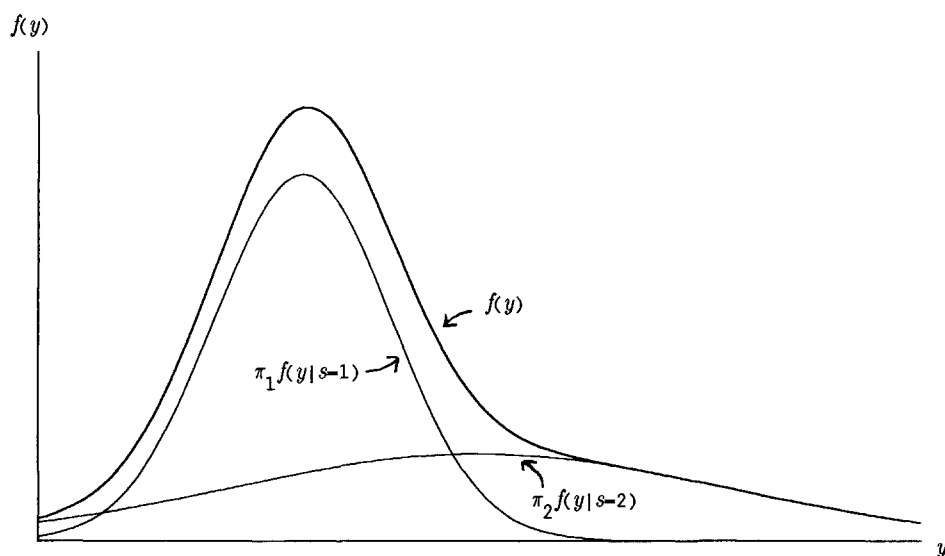


FIGURE 22.3 Density of mixture of two Gaussian distributions with $y_t|s_t = 1 \sim N(0, 1)$, $y_t|s_t = 2 \sim N(2, 8)$, and $P\{s_t = 1\} = 0.6$.

date t observation of y_t . Again, from the definition of a conditional probability, it follows that

$$P\{s_t = j | y_t; \boldsymbol{\theta}\} = \frac{p(y_t, s_t = j; \boldsymbol{\theta})}{f(y_t; \boldsymbol{\theta})} = \frac{\pi_j \cdot f(y_t | s_t = j; \boldsymbol{\theta})}{f(y_t; \boldsymbol{\theta})}. \quad [22.3.7]$$

Given knowledge of the population parameters $\boldsymbol{\theta}$, it would be possible to use [22.3.1] and [22.3.5] to calculate the magnitude in [22.3.7] for each observation y_t in the sample. This number represents the probability, given the observed data, that the unobserved regime responsible for observation t was regime j . For example, for the mixture represented in Figure 22.2, if an observation y_t were equal to zero, one could be virtually certain that the observation had come from a $N(0, 1)$ distribution rather than a $N(4, 1)$ distribution, so that $P\{s_t = 1 | y_t; \boldsymbol{\theta}\}$ for that date would be near unity. If instead y_t were around 2.3, it is equally likely that the observation might have come from either regime, so that $P\{s_t = 1 | y_t; \boldsymbol{\theta}\}$ for such an observation would be close to 0.5.

Maximum Likelihood Estimates and the EM Algorithm

It is instructive to characterize analytically the maximum likelihood estimates of the population parameter $\boldsymbol{\theta}$. Appendix 22.A demonstrates that the maximum likelihood estimate $\hat{\boldsymbol{\theta}}$ represents a solution to the following system of nonlinear equations:

$$\hat{\mu}_j = \frac{\sum_{t=1}^T y_t \cdot P\{s_t = j | y_t; \hat{\boldsymbol{\theta}}\}}{\sum_{t=1}^T P\{s_t = j | y_t; \hat{\boldsymbol{\theta}}\}} \quad \text{for } j = 1, 2, \dots, N \quad [22.3.8]$$

$$\hat{\sigma}_j^2 = \frac{\sum_{t=1}^T (y_t - \hat{\mu}_j)^2 \cdot P\{s_t = j | y_t; \hat{\boldsymbol{\theta}}\}}{\sum_{t=1}^T P\{s_t = j | y_t; \hat{\boldsymbol{\theta}}\}} \quad \text{for } j = 1, 2, \dots, N \quad [22.3.9]$$

$$\hat{\pi}_j = T^{-1} \sum_{t=1}^T P\{s_t = j | y_t; \hat{\boldsymbol{\theta}}\} \quad \text{for } j = 1, 2, \dots, N. \quad [22.3.10]$$

Suppose we were virtually certain which observations came from regime j and which did not, so that $P\{s_t = j | y_t; \boldsymbol{\theta}\}$ equaled unity for those observations that came from regime j and equaled zero for those observations that came from other regimes. Then the estimate of the mean for regime j in [22.3.8] would simply be the average value of y_t for those observations known to have come from regime j . In the more general case where $P\{s_t = j | y_t; \boldsymbol{\theta}\}$ is between 0 and 1 for some observations, the estimate $\hat{\mu}_j$ is a weighted average of all the observations in the sample, where the weight for observation y_t is proportional to the probability that date t 's observation was generated by regime j . The more likely an observation is to have come from regime j , the bigger the weight given that observation in estimating μ_j . Similarly, $\hat{\sigma}_j^2$ is a weighted average of the squared deviations of y_t from $\hat{\mu}_j$, while $\hat{\pi}_j$ is essentially the fraction of observations that appear to have come from regime j .

Because equations [22.3.8] to [22.3.10] are nonlinear, it is not possible to solve them analytically for $\hat{\boldsymbol{\theta}}$ as a function of $\{y_1, y_2, \dots, y_T\}$. However, these equations do suggest an appealing iterative algorithm for finding the maximum

likelihood estimate. Starting from an arbitrary initial guess for the value of θ , denoted $\theta^{(0)}$, one could calculate $P\{s_i = j|y_i; \theta^{(0)}\}$ from [22.3.7]. One could then calculate the magnitudes on the right sides of [22.3.8] through [22.3.10] with $\theta^{(0)}$ in place of $\hat{\theta}$. The left sides of [22.3.8] through [22.3.10] would then produce a new estimate $\theta^{(1)}$. This estimate $\theta^{(1)}$ could be used to reevaluate $P\{s_i = j|y_i; \theta^{(1)}\}$ and recalculate the expressions on the right sides of [22.3.8] through [22.3.10]. The left sides of [22.3.8] through [22.3.10] then can produce a new estimate $\theta^{(2)}$. One continues iterating in this fashion until the change between $\theta^{(m+1)}$ and $\theta^{(m)}$ is smaller than some specified convergence criterion.

This algorithm turns out to be a special case of the *EM* principle developed by Dempster, Laird, and Rubin (1977). One can show that each iteration on this algorithm increases the value of the likelihood function. Clearly, if the iterations reach a point such that $\theta^{(m)} = \theta^{(m+1)}$, the algorithm has found the maximum likelihood estimate $\hat{\theta}$.

Further Discussion

The mixture density [22.3.5] has the property that a global maximum of the log likelihood [22.3.6] does not exist. A singularity arises whenever one of the distributions is imputed to have a mean exactly equal to one of the observations ($\mu_1 = y_1$, say) with no variance ($\sigma_1^2 \rightarrow 0$). At such a point the log likelihood becomes infinite.

Such singularities do not pose a major problem in practice, since numerical maximization procedures typically converge to a reasonable local maximum rather than a singularity. The largest local maximum with $\sigma_j > 0$ for all j is described as the *maximum likelihood estimate*. Kiefer (1978) showed that there exists a bounded local maximum of [22.3.6] that yields a consistent, asymptotically Gaussian estimate of θ for which standard errors can be constructed using the usual formulas such as expression [5.8.3]. Hence, if a numerical maximization algorithm becomes stuck at a singularity, one satisfactory solution is simply to ignore the singularity and try again with different starting values.

Another approach is to maximize a slightly different objective function such as

$$Q(\theta) = \mathcal{L}(\theta) - \sum_{j=1}^N (a_j/2) \log(\sigma_j^2) - \sum_{j=1}^N b_j/(2\sigma_j^2) - \sum_{j=1}^N c_j(m_j - \mu_j)^2/(2\sigma_j^2), \quad [22.3.11]$$

where $\mathcal{L}(\theta)$ is the log likelihood function described in [22.3.6]. If $a_j = c_j$, then expression [22.3.11] is the form the log likelihood would take if, in addition to the data, the analyst had a_j observations from regime j whose sample mean was m_j and whose sample variance was b_j/a_j . Thus, m_j represents the analyst's prior expectation of the value of μ_j , and b_j/a_j represents the analyst's prior expectation of the value of σ_j^2 . The parameters a_j and c_j represent the strength of these priors, measured in terms of the confidence one would have if the priors were based on a_j or c_j direct observations of data known to have come from regime j . See Hamilton (1991) for further discussion of this approach.

Nice surveys of i.i.d. mixture distributions have been provided by Everitt and Hand (1981) and Titterton, Smith, and Makov (1985).

22.4. Time Series Models of Changes in Regime

Description of the Process

We now return to the objective of developing a model that allows a given variable to follow a different time series process over different subsamples. As an illustration, consider a first-order autoregression in which both the constant term and the autoregressive coefficient might be different for different subsamples:

$$y_t = c_{s_t} + \phi_{s_t} y_{t-1} + \varepsilon_t, \quad [22.4.1]$$

where $\varepsilon_t \sim \text{i.i.d. } N(0, \sigma^2)$. The proposal will be to model the regime s_t as the outcome of an unobserved N -state Markov chain with s_t independent of ε_τ for all t and τ .

Why might a Markov chain be a useful description of the process generating changes in regime? One's first thought could be that a change in regime such as that in Figure 22.1 is a permanent event. Such a permanent regime change could be modeled with a two-state Markov chain in which state 2 is an absorbing state. The advantage of using a Markov chain over a deterministic specification for such a process is that it allows one to generate meaningful forecasts prior to the change that take into account the possibility of the change from regime 1 to regime 2.

We might also want a time series model of changes in regime to account for unusual short-lived events such as World War II. Again, it is possible to choose parameters for a Markov chain such that, given 100 years of data, it is quite likely that we would have observed a single episode of regime 2 lasting for about 5 years. A Markov chain specification, of course, implies that given another 100 years we could well see another such event. One might argue that this is a sensible property to build into a model. The essence of the scientific method is the presumption that the future will in some sense be like the past.

While the Markov chain can describe such examples of changes in regime, a further advantage is its flexibility. There seems some value in specifying a probability law consistent with a broad range of different outcomes, and choosing particular parameters within that class on the basis of the data alone.

In any case, the approach described here readily generalizes to processes in which the probability that $s_t = j$ depends not only on the value of s_{t-1} but also on a vector of other observed variables—see Filardo (1994) and Diebold, Lee, and Weinbach (forthcoming).

The general model investigated in this section is as follows. Let $\mathbf{y}_t$ be an $(n \times 1)$ vector of observed endogenous variables and $\mathbf{x}_t$ a $(k \times 1)$ vector of observed exogenous variables. Let $\mathbf{y}_t = (\mathbf{y}'_t, \mathbf{y}'_{t-1}, \dots, \mathbf{y}'_{t-m}, \mathbf{x}'_t, \mathbf{x}'_{t-1}, \dots, \mathbf{x}'_{t-m})'$ be a vector containing all observations obtained through date t . If the process is governed by regime $s_t = j$ at date t , then the conditional density of $\mathbf{y}_t$ is assumed to be given by

$$f(\mathbf{y}_t | s_t = j, \mathbf{x}_t, \mathbf{y}_{t-1}; \boldsymbol{\alpha}), \quad [22.4.2]$$

where $\boldsymbol{\alpha}$ is a vector of parameters characterizing the conditional density. If there are N different regimes, then there are N different densities represented by [22.4.2] for $j = 1, 2, \dots, N$. These densities will be collected in an $(N \times 1)$ vector denoted $\boldsymbol{\eta}_t$.

For the example of [22.4.1], y_t is a scalar ($n = 1$), the exogenous variables consist only of a constant term ($\mathbf{x}_t = 1$), and the unknown parameters in $\boldsymbol{\alpha}$ consist of $c_1, \dots, c_N, \phi_1, \dots, \phi_N$, and σ^2 . With $N = 2$ regimes the two densities

represented by [22.4.2] are

$$\boldsymbol{\eta}_t = \begin{bmatrix} f(y_t | s_t = 1, y_{t-1}; \boldsymbol{\alpha}) \\ f(y_t | s_t = 2, y_{t-1}; \boldsymbol{\alpha}) \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(y_t - c_1 - \phi_1 y_{t-1})^2}{2\sigma^2}\right\} \\ \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(y_t - c_2 - \phi_2 y_{t-1})^2}{2\sigma^2}\right\} \end{bmatrix}.$$

It is assumed in [22.4.2] that the conditional density depends only on the current regime s_t and not on past regimes:

$$f(y_t | \mathbf{x}_t, \mathbf{y}_{t-1}, s_t = j; \boldsymbol{\alpha}) = f(y_t | \mathbf{x}_t, \mathbf{y}_{t-1}, s_t = j, s_{t-1} = i, s_{t-2} = k, \dots; \boldsymbol{\alpha}), \quad [22.4.3]$$

though this is not really restrictive. Consider, for example, the specification in [22.1.3], where the conditional density of y_t depends on both s_t^* and s_{t-1}^* and where s_t^* is described by a two-state Markov chain. One can define a new variable s_t that characterizes the regime for date t in a way consistent with [22.4.2] as follows:

$$\begin{aligned} s_t = 1 & \quad \text{if } s_t^* = 1 \text{ and } s_{t-1}^* = 1 \\ s_t = 2 & \quad \text{if } s_t^* = 2 \text{ and } s_{t-1}^* = 1 \\ s_t = 3 & \quad \text{if } s_t^* = 1 \text{ and } s_{t-1}^* = 2 \\ s_t = 4 & \quad \text{if } s_t^* = 2 \text{ and } s_{t-1}^* = 2. \end{aligned}$$

If p_{ij}^* denotes $P\{s_t^* = j | s_{t-1}^* = i\}$, then s_t follows a four-state Markov chain with transition matrix

$$\mathbf{P} = \begin{bmatrix} p_{11}^* & 0 & p_{11}^* & 0 \\ p_{12}^* & 0 & p_{12}^* & 0 \\ 0 & p_{21}^* & 0 & p_{21}^* \\ 0 & p_{22}^* & 0 & p_{22}^* \end{bmatrix}.$$

Hence, [22.1.3] could be represented as a special case of this framework with $N = 4$, $\boldsymbol{\alpha} = (\mu_1, \mu_2, \phi, \sigma^2)'$ and with [22.4.2] representing the four densities

$$\begin{aligned} f(y_t | y_{t-1}, s_t = 1; \boldsymbol{\alpha}) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{[(y_t - \mu_1) - \phi(y_{t-1} - \mu_1)]^2}{2\sigma^2}\right\} \\ f(y_t | y_{t-1}, s_t = 2; \boldsymbol{\alpha}) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{[(y_t - \mu_2) - \phi(y_{t-1} - \mu_1)]^2}{2\sigma^2}\right\} \\ f(y_t | y_{t-1}, s_t = 3; \boldsymbol{\alpha}) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{[(y_t - \mu_1) - \phi(y_{t-1} - \mu_2)]^2}{2\sigma^2}\right\} \\ f(y_t | y_{t-1}, s_t = 4; \boldsymbol{\alpha}) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{[(y_t - \mu_2) - \phi(y_{t-1} - \mu_2)]^2}{2\sigma^2}\right\}. \end{aligned}$$

It is assumed that s_t evolves according to a Markov chain that is independent of past observations on $\mathbf{y}_t$ or current or past $\mathbf{x}_t$:

$$P\{s_t = j | s_{t-1} = i, s_{t-2} = k, \dots, \mathbf{x}_t, \mathbf{y}_{t-1}\} = P\{s_t = j | s_{t-1} = i\} = p_{ij}. \quad [22.4.4]$$

For generalizations of this assumption, see Lam (1990), Durland and McCurdy (1994), Filardo (1994), and Diebold, Lee, and Weinbach (forthcoming).

Optimal Inference About Regimes and Evaluation of the Likelihood Function

The population parameters that describe a time series governed by [22.4.2] and [22.4.4] consist of α and the various transition probabilities p_{ij} . Collect these parameters in a vector θ . One important objective will be to estimate the value of θ based on observation of $\mathbf{y}_T$. Let us nevertheless put this objective on hold for the moment and suppose that the value of θ is somehow known with certainty to the analyst. Even if we know the value of θ , we will not know which regime the process was in at every date in the sample. Instead the best we can do is to form a probabilistic inference that is a generalization of [22.3.7]. In the i.i.d. case, the analyst's inference about the value of s_t depends only on the value of y_t . In the more general class of time series models described here the inference typically depends on all the observations available.

Let $P\{s_t = j | \mathbf{y}_t; \theta\}$ denote the analyst's inference about the value of s_t based on data obtained through date t and based on knowledge of the population parameters θ . This inference takes the form of a conditional probability that the analyst assigns to the possibility that the t th observation was generated by regime j . Collect these conditional probabilities $P\{s_t = j | \mathbf{y}_t; \theta\}$ for $j = 1, 2, \dots, N$ in an $(N \times 1)$ vector denoted $\hat{\xi}_{t|t}$.

One could also imagine forming forecasts of how likely the process is to be in regime j in period $t + 1$ given observations obtained through date t . Collect these forecasts in an $(N \times 1)$ vector $\hat{\xi}_{t+1|t}$, which is a vector whose j th element represents $P\{s_{t+1} = j | \mathbf{y}_t; \theta\}$.

The optimal inference and forecast for each date t in the sample can be found by iterating on the following pair of equations:

$$\hat{\xi}_{t|t} = \frac{(\hat{\xi}_{t|t-1} \odot \boldsymbol{\eta}_t)}{\mathbf{1}'(\hat{\xi}_{t|t-1} \odot \boldsymbol{\eta}_t)} \quad [22.4.5]$$

$$\hat{\xi}_{t+1|t} = \mathbf{P} \cdot \hat{\xi}_{t|t}. \quad [22.4.6]$$

Here $\boldsymbol{\eta}_t$ represents the $(N \times 1)$ vector whose j th element is the conditional density in [22.4.2], $\mathbf{P}$ represents the $(N \times N)$ transition matrix defined in [22.2.3], $\mathbf{1}$ represents an $(N \times 1)$ vector of 1s, and the symbol $\odot$ denotes element-by-element multiplication. Given a starting value $\hat{\xi}_{1|0}$ and an assumed value for the population parameter vector θ , one can iterate on [22.4.5] and [22.4.6] for $t = 1, 2, \dots, T$ to calculate the values of $\hat{\xi}_{t|t}$ and $\hat{\xi}_{t+1|t}$ for each date t in the sample. The log likelihood function $\mathcal{L}(\theta)$ for the observed data $\mathbf{y}_T$ evaluated at the value of θ that was used to perform the iterations can also be calculated as a by-product of this algorithm from

$$\mathcal{L}(\theta) = \sum_{t=1}^T \log f(\mathbf{y}_t | \mathbf{x}_t, \mathbf{y}_{t-1}; \theta), \quad [22.4.7]$$

where

$$f(\mathbf{y}_t | \mathbf{x}_t, \mathbf{y}_{t-1}; \theta) = \mathbf{1}'(\hat{\xi}_{t|t-1} \odot \boldsymbol{\eta}_t). \quad [22.4.8]$$

We now explain why this algorithm works.

Derivation of Equations [22.4.5] Through [22.4.8]

To see the basis for the algorithm just described, note that we have assumed that $\mathbf{x}_t$ is exogenous, by which we mean that $\mathbf{x}_t$ contains no information about s_t ,

beyond that contained in $\mathcal{Y}_{t-1}$. Hence, the j th element of $\hat{\xi}_{t|t-1}$ could also be described as $P\{s_t = j | \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta}\}$. The j th element of $\boldsymbol{\eta}_t$ is $f(\mathbf{y}_t | s_t = j, \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta})$. The j th element of the $(N \times 1)$ vector $(\hat{\xi}_{t|t-1} \odot \boldsymbol{\eta}_t)$ is the product of these two magnitudes, which product can be interpreted as the conditional joint density-distribution of $\mathbf{y}_t$ and s_t :

$$\begin{aligned} P\{s_t = j | \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta}\} \times f(\mathbf{y}_t | s_t = j, \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta}) \\ = p(\mathbf{y}_t, s_t = j | \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta}). \end{aligned} \quad [22.4.9]$$

The density of the observed vector $\mathbf{y}_t$ conditioned on past observables is the sum of the N magnitudes in [22.4.9] for $j = 1, 2, \dots, N$. This sum can be written in vector notation as

$$f(\mathbf{y}_t | \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta}) = \mathbf{1}'(\hat{\xi}_{t|t-1} \odot \boldsymbol{\eta}_t),$$

as claimed in [22.4.8]. If the joint density-distribution in [22.4.9] is divided by the density of $\mathbf{y}_t$ in [22.4.8], the result is the conditional distribution of s_t :

$$\begin{aligned} \frac{p(\mathbf{y}_t, s_t = j | \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta})}{f(\mathbf{y}_t | \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta})} &= P\{s_t = j | \mathbf{y}_t, \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta}\} \\ &= P\{s_t = j | \mathcal{Y}_t; \boldsymbol{\theta}\}. \end{aligned}$$

Hence, from [22.4.8],

$$P\{s_t = j | \mathcal{Y}_t; \boldsymbol{\theta}\} = \frac{p(\mathbf{y}_t, s_t = j | \mathbf{x}_t, \mathcal{Y}_{t-1}; \boldsymbol{\theta})}{\mathbf{1}'(\hat{\xi}_{t|t-1} \odot \boldsymbol{\eta}_t)}. \quad [22.4.10]$$

But recall from [22.4.9] that the numerator in the expression on the right side of [22.4.10] is the j th element of the vector $(\hat{\xi}_{t|t-1} \odot \boldsymbol{\eta}_t)$, while the left side of [22.4.10] is the j th element of the vector $\hat{\xi}_{t|t}$. Thus, collecting the equations in [22.4.10] for $j = 1, 2, \dots, N$ into an $(N \times 1)$ vector produces

$$\hat{\xi}_{t|t} = \frac{(\hat{\xi}_{t|t-1} \odot \boldsymbol{\eta}_t)}{\mathbf{1}'(\hat{\xi}_{t|t-1} \odot \boldsymbol{\eta}_t)},$$

as claimed in [22.4.5].

To see the basis for [22.4.6], take expectations of [22.2.6] conditional on $\mathcal{Y}_t$:

$$E(\xi_{t+1} | \mathcal{Y}_t) = \mathbf{P} \cdot E(\xi_t | \mathcal{Y}_t) + E(\mathbf{v}_{t+1} | \mathcal{Y}_t). \quad [22.4.11]$$

Note that $\mathbf{v}_{t+1}$ is a martingale difference sequence with respect to $\mathcal{Y}_t$, so that [22.4.11] becomes

$$\hat{\xi}_{t+1|t} = \mathbf{P} \cdot \hat{\xi}_{t|t},$$

as claimed in [22.4.6].

Starting the Algorithm

Given a starting value $\hat{\xi}_{1|0}$, one can use [22.4.5] and [22.4.6] to calculate $\hat{\xi}_{t|t}$ for any t . Several options are available for choosing the starting value. One approach is to set $\hat{\xi}_{1|0}$ equal to the vector of unconditional probabilities $\boldsymbol{\pi}$ described in equation [22.2.26]. Another option is to set

$$\hat{\xi}_{1|0} = \boldsymbol{\rho}, \quad [22.4.12]$$

where $\mathbf{p}$ is a fixed $(N \times 1)$ vector of nonnegative constants summing to unity, such as $\mathbf{p} = N^{-1} \cdot \mathbf{1}$. Alternatively, $\mathbf{p}$ could be estimated by maximum likelihood along with $\boldsymbol{\theta}$ subject to the constraint that $\mathbf{1}'\mathbf{p} = 1$ and $p_j \geq 0$ for $j = 1, 2, \dots, N$.

Forecasts and Smoothed Inferences for the Regime

Generalizing the earlier notation, let $\hat{\xi}_{t|\tau}$ represent the $(N \times 1)$ vector whose j th element is $P\{s_t = j | \mathcal{Y}_\tau; \boldsymbol{\theta}\}$. For $t > \tau$, this represents a forecast about the regime for some future period, whereas for $t < \tau$ it represents the *smoothed inference* about the regime the process was in at date t based on data obtained through some later date τ .

The optimal m -period-ahead forecast of ξ_{t+m} can be found by taking expectations of both sides of [22.2.8] conditional on information available at date t :

$$E(\xi_{t+m} | \mathcal{Y}_t) = \mathbf{P}^m \cdot E(\xi_t | \mathcal{Y}_t)$$

or

$$\hat{\xi}_{t+m|t} = \mathbf{P}^m \cdot \hat{\xi}_{t|t}, \quad [22.4.13]$$

where $\hat{\xi}_{t|t}$ is calculated from [22.4.5].

Smoothed inferences can be calculated using an algorithm developed by Kim (1993). In vector form, this algorithm can be written as

$$\hat{\xi}_{t|T} = \hat{\xi}_{t|t} \odot \{\mathbf{P}' \cdot [\hat{\xi}_{t+1|T} (\div) \hat{\xi}_{t+1|t}]\}, \quad [22.4.14]$$

where the sign $(\div)$ denotes element-by-element division. The smoothed probabilities $\hat{\xi}_{t|T}$ are found by iterating on [22.4.14] backward for $t = T - 1, T - 2, \dots, 1$. This iteration is started with $\hat{\xi}_{T|T}$, which is obtained from [22.4.5] for $t = T$. This algorithm is valid only when s_t follows a first-order Markov chain as in [22.4.4], when the conditional density [22.4.2] depends on $s_t, s_{t-1}, \dots$ only through the current state s_t , and when $\mathbf{x}_t$, the vector of explanatory variables other than the lagged values of $\mathbf{y}$, is strictly exogenous, meaning that $\mathbf{x}_t$ is independent of s_τ for all t and τ . The basis for Kim's algorithm is explained in Appendix 22.A at the end of the chapter.

Forecasts for the Observed Variables

From the conditional density [22.4.2] it is straightforward to forecast $\mathbf{y}_{t+1}$ conditional on knowing $\mathcal{Y}_t, \mathbf{x}_{t+1}$, and s_{t+1} . For example, for the $AR(1)$ specification $y_{t+1} = c_{s_{t+1}} + \phi_{s_{t+1}} y_t + \varepsilon_{t+1}$, such a forecast is given by

$$E(y_{t+1} | s_{t+1} = j, \mathcal{Y}_t; \boldsymbol{\theta}) = c_j + \phi_j y_t. \quad [22.4.15]$$

There are N different conditional forecasts associated with the N possible values for s_{t+1} . Note that the unconditional forecast based on actual observable variables is related to these conditional forecasts by

$$\begin{aligned} E(\mathbf{y}_{t+1} | \mathbf{x}_{t+1}, \mathcal{Y}_t; \boldsymbol{\theta}) &= \int \mathbf{y}_{t+1} \cdot f(\mathbf{y}_{t+1} | \mathbf{x}_{t+1}, \mathcal{Y}_t; \boldsymbol{\theta}) d\mathbf{y}_{t+1} \\ &= \int \mathbf{y}_{t+1} \left\{ \sum_{j=1}^N p(\mathbf{y}_{t+1}, s_{t+1} = j | \mathbf{x}_{t+1}, \mathcal{Y}_t; \boldsymbol{\theta}) \right\} d\mathbf{y}_{t+1} \end{aligned}$$

$$\begin{aligned}
&= \int \mathbf{y}_{t+1} \left\{ \sum_{j=1}^N [f(\mathbf{y}_{t+1} | s_{t+1} = j, \mathbf{x}_{t+1}, \mathbf{y}_t; \boldsymbol{\theta}) P\{s_{t+1} = j | \mathbf{x}_{t+1}, \mathbf{y}_t; \boldsymbol{\theta}\}] \right\} d\mathbf{y}_{t+1} \\
&= \sum_{j=1}^N P\{s_{t+1} = j | \mathbf{x}_{t+1}, \mathbf{y}_t; \boldsymbol{\theta}\} \int \mathbf{y}_{t+1} \cdot f(\mathbf{y}_{t+1} | s_{t+1} = j, \mathbf{x}_{t+1}, \mathbf{y}_t; \boldsymbol{\theta}) d\mathbf{y}_{t+1} \\
&= \sum_{j=1}^N P\{s_{t+1} = j | \mathbf{y}_t; \boldsymbol{\theta}\} E(\mathbf{y}_{t+1} | s_{t+1} = j, \mathbf{x}_{t+1}, \mathbf{y}_t; \boldsymbol{\theta}).
\end{aligned}$$

Thus, the forecast appropriate for the j th regime is simply multiplied by the probability that the process will be in the j th regime, and the resulting N different products are added together. For example, if the $j = 1, 2, \dots, N$ forecasts in [22.4.15] are collected in a $(1 \times N)$ vector $\mathbf{h}'_t$, then

$$E(\mathbf{y}_{t+1} | \mathbf{y}_t; \boldsymbol{\theta}) = \mathbf{h}'_t \hat{\boldsymbol{\xi}}_{t+1|t}.$$

Note that although the Markov chain itself admits the linear representation [22.2.6], the optimal forecast of $\mathbf{y}_{t+1}$ is a nonlinear function of observables, since the inference $\hat{\boldsymbol{\xi}}_{t|t}$ in [22.4.5] depends nonlinearly on $\mathbf{y}_t$. Although one may use a linear model to form forecasts within a given regime, if an observation seems unlikely to have been generated by the same regime as preceding observations, the appearance of the outlier causes the analyst to switch to a new rule for forming future linear forecasts.

The Markov chain is clearly well suited for forming multiperiod forecasts as well. See Hamilton (1989, 1993a, 1993b) for further discussion.

Maximum Likelihood Estimation of Parameters

In the iteration on [22.4.5] and [22.4.6], the parameter vector $\boldsymbol{\theta}$ was taken to be a fixed, known vector. Once the iteration has been completed for $t = 1, 2, \dots, T$ for a given fixed $\boldsymbol{\theta}$, the value of the log likelihood implied by that value of $\boldsymbol{\theta}$ is then known from [22.4.7]. The value of $\boldsymbol{\theta}$ that maximizes the log likelihood can be found numerically using the methods described in Section 5.7.

If the transition probabilities are restricted only by the conditions that $p_{ij} \geq 0$ and $(p_{i1} + p_{i2} + \dots + p_{iN}) = 1$ for all i and j , and if the initial probability $\hat{\boldsymbol{\xi}}_{1|0}$ is taken to be a fixed value $\boldsymbol{\rho}$ unrelated to the other parameters, then it is shown in Hamilton (1990) that the maximum likelihood estimates for the transition probabilities satisfy

$$\hat{p}_{ij} = \frac{\sum_{t=2}^T P\{s_t = j, s_{t-1} = i | \mathbf{y}_T; \hat{\boldsymbol{\theta}}\}}{\sum_{t=2}^T P\{s_{t-1} = i | \mathbf{y}_T; \hat{\boldsymbol{\theta}}\}}, \quad [22.4.16]$$

where $\hat{\boldsymbol{\theta}}$ denotes the full vector of maximum likelihood estimates. Thus, the estimated transition probability $\hat{p}_{ij}$ is essentially the number of times state i seems to have been followed by state j divided by the number of times the process was in state i . These counts are estimated on the basis of the smoothed probabilities.

If the vector of initial probabilities $\boldsymbol{\rho}$ is regarded as a separate vector of parameters constrained only by $\mathbf{1}'\boldsymbol{\rho} = 1$ and $\boldsymbol{\rho} \geq \mathbf{0}$, the maximum likelihood estimate of $\boldsymbol{\rho}$ turns out to be the smoothed inference about the initial state:

$$\hat{\boldsymbol{\rho}} = \hat{\boldsymbol{\xi}}_{1|T}. \quad [22.4.17]$$

The maximum likelihood estimate of the vector α that governs the conditional density [22.4.2] is characterized by

$$\sum_{t=1}^T \left(\frac{\partial \log \eta_t}{\partial \alpha'} \right)' \hat{\xi}_{t|T} = \mathbf{0}. \quad [22.4.18]$$

Here η_t is the $(N \times 1)$ vector obtained by vertically stacking the densities in [22.4.2] for $j = 1, 2, \dots, N$ and $(\partial \log \eta_t)/\partial \alpha'$ is the $(N \times k)$ matrix of derivatives of the logs of these densities, where k represents the number of parameters in α . For example, consider a Markov-switching regression model of the form

$$y_t = \mathbf{z}_t' \boldsymbol{\beta}_{s_t} + \varepsilon_t, \quad [22.4.19]$$

where $\varepsilon_t \sim \text{i.i.d. } N(0, \sigma^2)$ and where $\mathbf{z}_t$ is a vector of explanatory variables that could include lagged values of y . The coefficient vector for this regression is $\boldsymbol{\beta}_1$ when the process is in regime 1, $\boldsymbol{\beta}_2$ when the process is in regime 2, and so on. For this example, the vector η_t would be

$$\eta_t = \begin{bmatrix} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{ \frac{-(y_t - \mathbf{z}_t' \boldsymbol{\beta}_1)^2}{2\sigma^2} \right\} \\ \vdots \\ \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{ \frac{-(y_t - \mathbf{z}_t' \boldsymbol{\beta}_N)^2}{2\sigma^2} \right\} \end{bmatrix},$$

and for $\alpha = (\boldsymbol{\beta}'_1, \boldsymbol{\beta}'_2, \dots, \boldsymbol{\beta}'_N, \sigma^2)'$, condition [22.4.18] becomes

$$\sum_{t=1}^T (y_t - \mathbf{z}_t' \hat{\boldsymbol{\beta}}_j) \mathbf{z}_t \cdot P\{s_t = j | \mathbf{y}_T; \hat{\boldsymbol{\theta}}\} = \mathbf{0} \quad \text{for } j = 1, 2, \dots, N \quad [22.4.20]$$

$$\hat{\sigma}^2 = T^{-1} \sum_{t=1}^T \sum_{j=1}^N (y_t - \mathbf{z}_t' \hat{\boldsymbol{\beta}}_j)^2 \cdot P\{s_t = j | \mathbf{y}_T; \hat{\boldsymbol{\theta}}\}. \quad [22.4.21]$$

Equation [22.4.20] describes $\hat{\boldsymbol{\beta}}_j$ as satisfying a weighted *OLS* orthogonality condition where each observation is weighted by the probability that it came from regime j . In particular, the estimate $\hat{\boldsymbol{\beta}}_j$ can be found from an *OLS* regression of $\tilde{y}_t(j)$ on $\tilde{\mathbf{z}}_t(j)$:

$$\hat{\boldsymbol{\beta}}_j = \left[\sum_{t=1}^T [\tilde{\mathbf{z}}_t(j)] [\tilde{\mathbf{z}}_t(j)]' \right]^{-1} \left[\sum_{t=1}^T [\tilde{\mathbf{z}}_t(j)] \tilde{y}_t(j) \right], \quad [22.4.22]$$

where

$$\begin{aligned} \tilde{y}_t(j) &= y_t \cdot \sqrt{P\{s_t = j | \mathbf{y}_T; \hat{\boldsymbol{\theta}}\}} \\ \tilde{\mathbf{z}}_t(j) &= \mathbf{z}_t \cdot \sqrt{P\{s_t = j | \mathbf{y}_T; \hat{\boldsymbol{\theta}}\}}. \end{aligned} \quad [22.4.23]$$

The estimate of σ^2 in [22.4.21] is just $(1/T)$ times the combined sum of the squared residuals from these N different regressions.

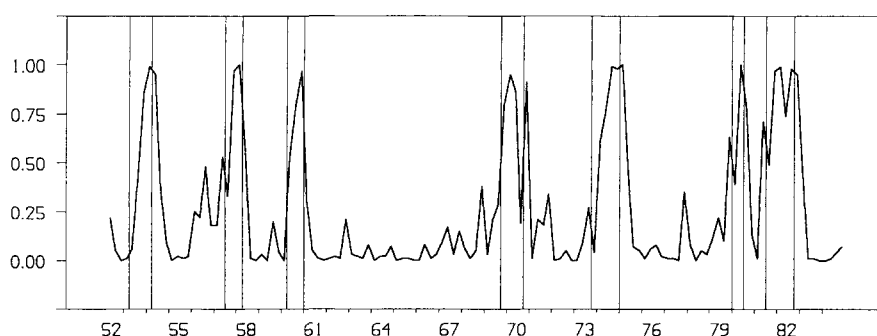
Again, this suggests an appealing algorithm for finding maximum likelihood estimates. For the case when $\boldsymbol{\rho}$ is fixed a priori, given an initial guess for the parameter vector $\boldsymbol{\theta}^{(0)}$ one could evaluate [22.4.16], [22.4.22], and [22.4.21] to generate a new estimate $\boldsymbol{\theta}^{(1)}$. One then iterates in the same fashion described in equations [22.3.8] through [22.3.10] to calculate $\boldsymbol{\theta}^{(2)}$, $\boldsymbol{\theta}^{(3)}$, $\dots$. This again turns out to be an application of the *EM* algorithm. Alternatively, if $\boldsymbol{\rho}$ is to be estimated by maximum likelihood, equation [22.4.17] would be added to the equations that are reevaluated with each iteration. See Hamilton (1990) for details.

Illustration: The Behavior of U.S. Real GNP

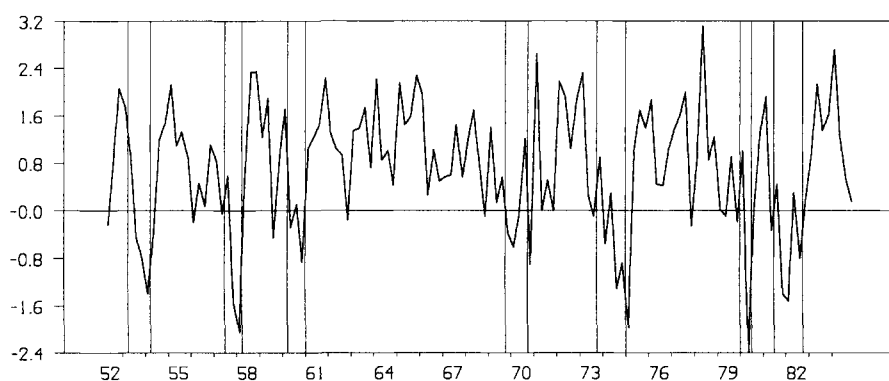
As an illustration of this method, consider the data on U.S. real GNP growth analyzed in Hamilton (1989). These data are plotted in the bottom panel of Figure 22.4. The following switching model was fitted to these data by maximum likelihood:

$$y_t - \mu_{s_t^*} = \phi_1(y_{t-1} - \mu_{s_{t-1}^*}) + \phi_2(y_{t-2} - \mu_{s_{t-2}^*}) + \phi_3(y_{t-3} - \mu_{s_{t-3}^*}) + \phi_4(y_{t-4} - \mu_{s_{t-4}^*}) + \varepsilon_t, \quad [22.4.24]$$

with $\varepsilon_t \sim \text{i.i.d. } N(0, \sigma^2)$ and with s_t^* presumed to follow a two-state Markov chain with transition probabilities p_{ij}^* . Maximum likelihood estimates of parameters are reported in Table 22.1. In the regime represented by $s_t^* = 1$, the average growth rate is $\mu_1 = 1.2\%$ per quarter, while when $s_t^* = 2$, the average growth rate is $\mu_2 = -0.4\%$. Each regime is highly persistent. The probability that expansion will be followed by another quarter of expansion is $p_{11}^* = 0.9$, so that this regime will persist on average for $1/(1 - p_{11}^*) = 10$ quarters. The probability that a contraction will be followed by contraction is $p_{22}^* = 0.75$, which episodes will typically persist for $1/(1 - p_{22}^*) = 4$ quarters.



(a) Probability that economy is in contraction state, or $P\{s_t^* = 2 | y_t, y_{t-1}, \dots, y_{t-4}; \hat{\theta}\}$ plotted as a function of t .



(b) Quarterly rate of growth of U.S. real GNP, 1952–84.

FIGURE 22.4 Output growth and recession probabilities.

TABLE 22.1
Maximum Likelihood Estimates of Parameters for Markov-Switching Model
of U.S. GNP (Standard Errors in Parentheses)

$\hat{\mu}_1 = 1.16$ (0.07)	$\hat{\mu}_2 = -0.36$ (0.26)	$\hat{\rho}_{11}^* = 0.90$ (0.04)	$\hat{\rho}_{22}^* = 0.75$ (0.10)	$\hat{\sigma}^2 = 0.59$ (0.10)
$\hat{\phi}_1 = 0.01$ (0.12)	$\hat{\phi}_2 = -0.06$ (0.14)	$\hat{\phi}_3 = -0.25$ (0.11)	$\hat{\phi}_4 = -0.21$ (0.11)	

In order to write [22.4.24] in a form where y_t depends only on the current value of the regime, a variable s_t was defined that takes on one of 32 different values representing the 32 possible combinations for $s_t^*, s_{t-1}^*, \dots, s_{t-4}^*$. For example, $s_t = 1$ when $s_t^*, s_{t-1}^*, \dots$, and s_{t-4}^* all equal 1, $s_t = 2$ when $s_t^* = 2$ and $s_{t-1}^* = \dots = s_{t-4}^* = 1$, and so on. The vector $\xi_{t|t}$ calculated from [22.4.5] is thus a (32×1) vector that contains the probabilities of each of these 32 joint events conditional on data observed through date t .

The inference about the value of s_t^* for a single date t is obtained by summing together the relevant joint probabilities. For example, the inference

$$\begin{aligned}
 P\{s_t^* = 2 | y_t, y_{t-1}, \dots, y_{-4}; \hat{\theta}\} \\
 = \sum_{i_1=1}^2 \sum_{i_2=1}^2 \sum_{i_3=1}^2 \sum_{i_4=1}^2 P\{s_t^* = 2, s_{t-1}^* = i_1, s_{t-2}^* = i_2, s_{t-3}^* = i_3, s_{t-4}^* = i_4 | y_t, \\
 y_{t-1}, \dots, y_{-4}; \hat{\theta}\}
 \end{aligned}
 \tag{22.4.25}$$

is obtained by iterating on [22.4.5] and [22.4.6] with θ equal to the maximum likelihood estimate $\hat{\theta}$. One then sums together the elements in the even-numbered rows of $\xi_{t|t}$ to obtain $P\{s_t^* = 2 | y_t, y_{t-1}, \dots, y_{-4}; \hat{\theta}\}$.

A probabilistic inference in the form of [22.4.25] can be calculated for each date t in the sample. The resulting series is plotted as a function of t in panel (a) of Figure 22.4. The vertical lines in the figure indicate the dates at which economic recessions were determined to begin and end according to the National Bureau of Economic Research. These determinations are made informally on the basis of a large number of time series and are usually made some time after the event. Although these business cycle dates were not used in any way to estimate parameters or form inferences about s_t^* , it is interesting that the traditional business cycle dates correspond fairly closely to the expansion and contraction phases as described by the model in [22.4.24].

Determining the Number of States

One of the most important hypotheses that one would want to test for such models concerns the number of different regimes N that characterize the data. Unfortunately, this hypothesis cannot be tested using the usual likelihood ratio test. One of the regularity conditions for the likelihood ratio test to have an asymptotic χ^2 distribution is that the information matrix $\mathcal{J}$ be nonsingular. This condition fails to hold if the analyst tries to fit an N -state model when the true process has $N - 1$ states, since under the null hypothesis the parameters that describe the N th state are unidentified. Tests that get around the problems with the regularity conditions have been proposed by Davies (1977), Hansen (1996), Andrews and Ploberger (1994), and Stinchcombe and White (1993). Another approach is to take

the $(N - 1)$ -state model as the null and conduct a variety of tests of the validity of that specification as one way of seeing whether an N -state model is needed; Hamilton (1996) proposed a number of such tests. Studies that illustrate the use of such tests include Engel and Hamilton (1990), Hansen (1992), and Goodwin (1993).

APPENDIX 22.A. Derivation of Selected Equations for Chapter 22

■ **Derivation of [22.3.8] through [22.3.10].** The maximum likelihood estimates are obtained by forming the Lagrangean

$$J(\theta) = \mathcal{L}(\theta) + \lambda(1 - \pi_1 - \pi_2 - \cdots - \pi_N) \quad [22.A.1]$$

and setting the derivative with respect to θ equal to zero. From [22.3.6], the derivative of the log likelihood is given by

$$\frac{\partial \mathcal{L}(\theta)}{\partial \theta} = \sum_{i=1}^T \frac{1}{f(y_i; \theta)} \times \frac{\partial f(y_i; \theta)}{\partial \theta}. \quad [22.A.2]$$

Observe from [22.3.5] that

$$\begin{aligned} \frac{\partial f(y_i; \theta)}{\partial \pi_j} &= \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left\{-\frac{(y_i - \mu_j)^2}{2\sigma_j^2}\right\} \\ &= f(y_i | s_i = j; \theta), \end{aligned} \quad [22.A.3]$$

while

$$\frac{\partial f(y_i; \theta)}{\partial \mu_j} = \frac{y_i - \mu_j}{\sigma_j^2} \times p(y_i, s_i = j; \theta) \quad [22.A.4]$$

and

$$\frac{\partial f(y_i; \theta)}{\partial \sigma_j^2} = \left\{ -\frac{1}{2} \sigma_j^{-2} + \frac{(y_i - \mu_j)^2}{2\sigma_j^4} \right\} \times p(y_i, s_i = j; \theta). \quad [22.A.5]$$

Thus, [22.A.2] becomes

$$\frac{\partial \mathcal{L}(\theta)}{\partial \pi_j} = \sum_{i=1}^T \frac{1}{f(y_i; \theta)} f(y_i | s_i = j; \theta) \quad [22.A.6]$$

$$\frac{\partial \mathcal{L}(\theta)}{\partial \mu_j} = \sum_{i=1}^T \frac{1}{f(y_i; \theta)} \times \frac{y_i - \mu_j}{\sigma_j^2} p(y_i, s_i = j; \theta) \quad [22.A.7]$$

$$\frac{\partial \mathcal{L}(\theta)}{\partial \sigma_j^2} = \sum_{i=1}^T \frac{1}{f(y_i; \theta)} \left\{ -\frac{1}{2} \sigma_j^{-2} + \frac{(y_i - \mu_j)^2}{2\sigma_j^4} \right\} p(y_i, s_i = j; \theta). \quad [22.A.8]$$

Recalling [22.3.7], the derivatives in [22.A.6] through [22.A.8] can be written

$$\frac{\partial \mathcal{L}(\theta)}{\partial \pi_j} = \pi_j^{-1} \sum_{i=1}^T P\{s_i = j | y_i; \theta\} \quad [22.A.9]$$

$$\frac{\partial \mathcal{L}(\theta)}{\partial \mu_j} = \sum_{i=1}^T \frac{y_i - \mu_j}{\sigma_j^2} P\{s_i = j | y_i; \theta\} \quad [22.A.10]$$

$$\frac{\partial \mathcal{L}(\theta)}{\partial \sigma_j^2} = \sum_{i=1}^T \left\{ -\frac{1}{2} \sigma_j^{-2} + \frac{(y_i - \mu_j)^2}{2\sigma_j^4} \right\} P\{s_i = j | y_i; \theta\}. \quad [22.A.11]$$

Setting the derivative of the Lagrangean in [22.A.1] with respect to μ_j equal to zero means setting [22.A.10] equal to zero, from which

$$\sum_{i=1}^T y_i \cdot P\{s_i = j | y_i; \theta\} = \mu_j \sum_{i=1}^T P\{s_i = j | y_i; \theta\}.$$

Equation [22.3.8] follows immediately from this condition. Similarly, the first-order conditions for maximization with respect to σ_j^2 are found by setting [22.A.11] equal to zero:

$$\sum_{t=1}^T \{-\sigma_j^2 + (y_t - \mu_j)^2\} P\{s_t = j | y_t; \theta\} = 0,$$

from which [22.3.9] follows. Finally, from [22.A.9], the derivative of [22.A.1] with respect to π_j is given by

$$\frac{\partial J(\theta)}{\partial \pi_j} = \pi_j^{-1} \sum_{t=1}^T P\{s_t = j | y_t; \theta\} - \lambda = 0,$$

from which

$$\sum_{t=1}^T P\{s_t = j | y_t; \theta\} = \lambda \pi_j. \quad [22.A.12]$$

Summing [22.A.12] over $j = 1, 2, \dots, N$ produces

$$\sum_{t=1}^T [P\{s_t = 1 | y_t; \theta\} + \dots + P\{s_t = N | y_t; \theta\}] = \lambda(\pi_1 + \pi_2 + \dots + \pi_N)$$

or

$$\sum_{t=1}^T \{1\} = \lambda \cdot (1),$$

implying that $T = \lambda$. Replacing λ with T in [22.A.12] produces [22.3.10]. ■

■ **Derivation of [22.4.14].** Recall first that under the maintained assumptions, the regime s_t depends on past observations $\mathbf{y}_{t-1}$ only through the value of s_{t-1} . Similarly, s_t depends on future observations only through the value of s_{t+1} :

$$P\{s_t = j | s_{t+1} = i, \mathbf{y}_T; \theta\} = P\{s_t = j | s_{t+1} = i, \mathbf{y}_t; \theta\}. \quad [22.A.13]$$

The validity of [22.A.13] is formally established as follows (the implicit dependence on θ will be suppressed to simplify the notation). Observe that

$$\begin{aligned} P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+1}\} &= P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+1}, \mathbf{x}_{t+1}, \mathbf{y}_t\} \\ &= \frac{p(\mathbf{y}_{t+1}, s_t = j | s_{t+1} = i, \mathbf{x}_{t+1}, \mathbf{y}_t)}{f(\mathbf{y}_{t+1} | s_{t+1} = i, \mathbf{x}_{t+1}, \mathbf{y}_t)} \\ &= \frac{f(\mathbf{y}_{t+1} | s_t = j, s_{t+1} = i, \mathbf{x}_{t+1}, \mathbf{y}_t) \cdot P\{s_t = j | s_{t+1} = i, \mathbf{x}_{t+1}, \mathbf{y}_t\}}{f(\mathbf{y}_{t+1} | s_{t+1} = i, \mathbf{x}_{t+1}, \mathbf{y}_t)}, \end{aligned} \quad [22.A.14]$$

which simplifies to

$$P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+1}\} = P\{s_t = j | s_{t+1} = i, \mathbf{x}_{t+1}, \mathbf{y}_t\}, \quad [22.A.15]$$

provided that

$$f(\mathbf{y}_{t+1} | s_t = j, s_{t+1} = i, \mathbf{x}_{t+1}, \mathbf{y}_t) = f(\mathbf{y}_{t+1} | s_{t+1} = i, \mathbf{x}_{t+1}, \mathbf{y}_t), \quad [22.A.16]$$

which is indeed the case, since the specification assumes that $\mathbf{y}_{t+1}$ depends on $\{s_{t+1}, s_t, \dots\}$ only through the current value s_{t+1} . Since $\mathbf{x}$ is exogenous, [22.A.15] further implies that

$$P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+1}\} = P\{s_t = j | s_{t+1} = i, \mathbf{y}_t\}. \quad [22.A.17]$$

By similar reasoning, it must be the case that

$$\begin{aligned} P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+2}\} &= P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+2}, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}\} \\ &= \frac{p(\mathbf{y}_{t+2}, s_t = j | s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1})}{f(\mathbf{y}_{t+2} | s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1})} \\ &= \frac{f(\mathbf{y}_{t+2} | s_t = j, s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}) \cdot P\{s_t = j | s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}\}}{f(\mathbf{y}_{t+2} | s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1})}, \end{aligned}$$

which simplifies to

$$P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+2}\} = P\{s_t = j | s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}\}, \quad [22.A.18]$$

provided that

$$f(\mathbf{y}_{t+2} | s_t = j, s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}) = f(\mathbf{y}_{t+2} | s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}). \quad [22.A.19]$$

In this case, [22.A.19] is established from the fact that

$$\begin{aligned} & f(\mathbf{y}_{t+2} | s_t = j, s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}) \\ &= \sum_{k=1}^N p(\mathbf{y}_{t+2}, s_{t+2} = k | s_t = j, s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}) \\ &= \sum_{k=1}^N [f(\mathbf{y}_{t+2} | s_{t+2} = k, s_t = j, s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}) \\ &\quad \times P\{s_{t+2} = k | s_t = j, s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}\}] \\ &= \sum_{k=1}^N [f(\mathbf{y}_{t+2} | s_{t+2} = k, s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}) \\ &\quad \times P\{s_{t+2} = k | s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}\}] \\ &= f(\mathbf{y}_{t+2} | s_{t+1} = i, \mathbf{x}_{t+2}, \mathbf{y}_{t+1}). \end{aligned}$$

Again, exogeneity of $\mathbf{x}$ means that [22.A.18] can be written

$$P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+2}\} = P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+1}\} = P\{s_t = j | s_{t+1} = i, \mathbf{y}_t\},$$

where the last equality follows from [22.A.17].

Proceeding inductively, the same argument can be used to establish that

$$P\{s_t = j | s_{t+1} = i, \mathbf{y}_{t+m}\} = P\{s_t = j | s_{t+1} = i, \mathbf{y}_t\}$$

for $m = 1, 2, \dots$, from which [22.A.13] follows.

Note next that

$$\begin{aligned} P\{s_t = j | s_{t+1} = i, \mathbf{y}_t\} &= \frac{P\{s_t = j, s_{t+1} = i | \mathbf{y}_t\}}{P\{s_{t+1} = i | \mathbf{y}_t\}} \\ &= \frac{P\{s_t = j | \mathbf{y}_t\} \cdot P\{s_{t+1} = i | s_t = j\}}{P\{s_{t+1} = i | \mathbf{y}_t\}} \\ &= \frac{p_{ji} \cdot P\{s_t = j | \mathbf{y}_t\}}{P\{s_{t+1} = i | \mathbf{y}_t\}}. \end{aligned} \quad [22.A.20]$$

It is therefore the case that

$$\begin{aligned} P\{s_t = j, s_{t+1} = i | \mathbf{y}_T\} &= P\{s_{t+1} = i | \mathbf{y}_T\} \cdot P\{s_t = j | s_{t+1} = i, \mathbf{y}_t\} \\ &= P\{s_{t+1} = i | \mathbf{y}_T\} \cdot P\{s_t = j | s_{t+1} = i, \mathbf{y}_t\} \\ &= P\{s_{t+1} = i | \mathbf{y}_T\} \frac{p_{ji} \cdot P\{s_t = j | \mathbf{y}_t\}}{P\{s_{t+1} = i | \mathbf{y}_t\}}, \end{aligned} \quad [22.A.21]$$

where the second equality follows from [22.A.13] and the third follows from [22.A.20].

The smoothed inference for date t is the sum of [22.A.21] over $i = 1, 2, \dots, N$:

$$\begin{aligned} P\{s_t = j | \mathbf{y}_T\} &= \sum_{i=1}^N P\{s_t = j, s_{t+1} = i | \mathbf{y}_T\} \\ &= \sum_{i=1}^N P\{s_{t+1} = i | \mathbf{y}_T\} \frac{p_{ji} \cdot P\{s_t = j | \mathbf{y}_t\}}{P\{s_{t+1} = i | \mathbf{y}_t\}} \\ &= P\{s_t = j | \mathbf{y}_t\} \sum_{i=1}^N \frac{p_{ji} \cdot P\{s_{t+1} = i | \mathbf{y}_T\}}{P\{s_{t+1} = i | \mathbf{y}_t\}} \\ &= P\{s_t = j | \mathbf{y}_t\} [p_{j1} \quad p_{j2} \quad \cdots \quad p_{jN}] \\ &\quad \times \begin{bmatrix} P\{s_{t+1} = 1 | \mathbf{y}_T\} / P\{s_{t+1} = 1 | \mathbf{y}_t\} \\ P\{s_{t+1} = 2 | \mathbf{y}_T\} / P\{s_{t+1} = 2 | \mathbf{y}_t\} \\ \vdots \\ P\{s_{t+1} = N | \mathbf{y}_T\} / P\{s_{t+1} = N | \mathbf{y}_t\} \end{bmatrix} \\ &= P\{s_t = j | \mathbf{y}_t\} \mathbf{p}'_j (\hat{\xi}_{t+1|T} \div) \hat{\xi}_{t+1|t}, \end{aligned} \quad [22.A.22]$$

where the $(1 \times N)$ vector $\mathbf{p}'_j$ denotes the j th row of the matrix $\mathbf{P}'$ and the sign $(\div)$ indicates element-by-element division. When the equations represented by [22.A.22] for $j = 1, 2, \dots, N$ are collected in an $(N \times 1)$ vector, the result is

$$\hat{\xi}_{t|T} = \hat{\xi}_{t|t} \odot \{\mathbf{P}'(\hat{\xi}_{t+1|T} \div \hat{\xi}_{t+1|t})\},$$

as claimed. ■

Chapter 22 Exercise

22.1. Let s_t be described by an ergodic two-state Markov chain with transition matrix $\mathbf{P}$ given by [22.2.11]. Verify that the matrix of eigenvectors of this matrix is given by

$$\mathbf{T} = \begin{bmatrix} (1 - p_{22})/(2 - p_{11} - p_{22}) & -1 \\ (1 - p_{11})/(2 - p_{11} - p_{22}) & 1 \end{bmatrix}$$

with inverse

$$\mathbf{T}^{-1} = \begin{bmatrix} 1 & 1 \\ -(1 - p_{11})/(2 - p_{11} - p_{22}) & (1 - p_{22})/(2 - p_{11} - p_{22}) \end{bmatrix}.$$

Chapter 22 References

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